Computer Networks - Project 2

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1 Exercise 1

1.1 Network Overview

The complete configuration is in the .pkt file, however we provide an overview here.

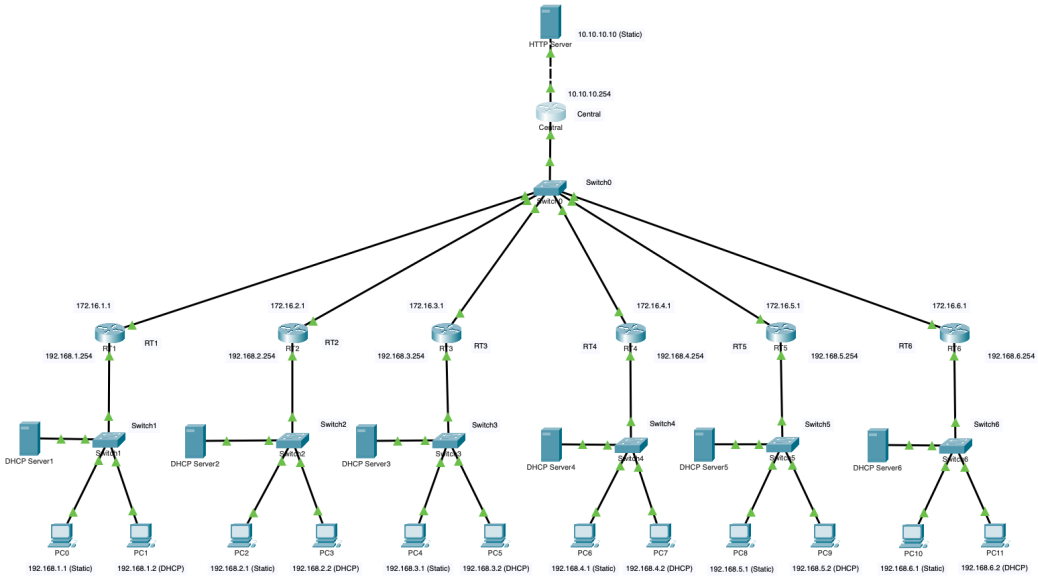


Figure 1: Topology for Exercise 1

1.2 OSPF Routing Configuration

1.2.1 Central Router

On the central router, OSPF process 1 is enabled. The configuration utilizes wildcard masks to efficiently advertise multiple subnets.

```
Router#enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 172.16.0.0 0.0.7.255 area 0
Router(config-router)#network 10.10.10.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 2: Central Router OSPF CLI Configuration

- **Core Network:** `172.16.0.0 0.0.7.255` area 0. This wildcard mask includes all /29 subnets from 172.16.1.0 through 172.16.6.0.
- **Server LAN:** `10.10.10.0 0.0.0.255` area 0. Advertises the HTTP server segment.

1.2.2 Branch Router Configurations (RT1–RT6)

Each branch router advertises its local LAN and its point-to-point link to the core.

```
RT1>enable
RT1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT1(config)#router ospf 1
RT1(config-router)#network 172.16.1.0 0.0.0.7 area 0
RT1(config-router)#network 192.168.1.0 0.0.0.255 area 0
RT1(config-router)#end
RT1#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 3: RT1 OSPF

```
RT2>enable
RT2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT2(config)#router ospf 1
RT2(config-router)#network 172.16.2.0 0.0.0.7 area 0
RT2(config-router)#network 192.168.2.0 0.0.0.255 area 0
RT2(config-router)#end
RT2#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 4: RT2 OSPF

```
RT3>enable
RT3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT3(config)#router ospf 1
RT3(config-router)#network 172.16.3.0 0.0.0.7 area 0
RT3(config-router)#network 192.168.3.0 0.0.0.255 area 0
RT3(config-router)#end
RT3#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 5: RT3 OSPF

```
RT4>enable
RT4#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT4(config)#router ospf 1
RT4(config-router)#network 172.16.4.0 0.0.0.7 area 0
RT4(config-router)#network 192.168.4.0 0.0.0.255 area 0
RT4(config-router)#end
RT4#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 6: RT4 OSPF

```
RT5>enable
RT5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT5(config)#router ospf 1
RT5(config-router)#network 172.16.5.0 0.0.0.7 area 0
RT5(config-router)#network 192.168.5.0 0.0.0.255 area 0
RT5(config-router)#end
RT5#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 7: RT5 OSPF

```
RT6>enable
RT6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RT6(config)#router ospf 1
RT6(config-router)#network 172.16.6.0 0.0.0.7 area 0
RT6(config-router)#network 192.168.6.0 0.0.0.255 area 0
RT6(config-router)#end
RT6#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 8: RT6 OSPF

1.3 OSPF Operational Explanation

OSPF (Open Shortest Path First) functions through several key stages to ensure loop-free routing:

- **Neighbor Discovery:** Routers send Hello packets every 10s. Adjacencies form only if Hello/Dead intervals and Area IDs match.
- **DR/BDR Election:** On the `172.16.x.0/29` segments, a Designated Router (DR) is elected to manage Link-State Advertisement (LSA) flooding, reducing network overhead.
- **Database Synchronization:** Routers exchange LSAs to build a synchronized Link-State Database (LSDB).
- **SPF Calculation:** Each router runs Dijkstra's algorithm to calculate the shortest path to all destination subnets.

```
CentralRT#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.1.254     1   FULL/DR         00:00:34    172.16.1.1      FastEthernet0/1.1
192.168.2.254     1   FULL/DR         00:00:34    172.16.2.1      FastEthernet0/1.2
192.168.3.254     1   FULL/DR         00:00:34    172.16.3.1      FastEthernet0/1.3
192.168.4.254     1   FULL/DR         00:00:34    172.16.4.1      FastEthernet0/1.4
192.168.5.254     1   FULL/DR         00:00:34    172.16.5.1      FastEthernet0/1.5
192.168.6.254     1   FULL/DR         00:00:36    172.16.6.1      FastEthernet0/1.6
CentralRT#
```

Figure 9: OSPF Neighbor Adjacency Table

```
CentralRT#show ip ospf database
            OSPF Router with ID (172.16.6.6) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum Link count
172.16.6.6      172.16.6.6      516         0x80000014 0x0023e0 7
192.168.4.254   192.168.4.254   521         0x8000000a 0x00145d 2
192.168.2.254   192.168.2.254   520         0x8000000a 0x00c9b1 2
192.168.3.254   192.168.3.254   516         0x8000000a 0x00ee87 2
192.168.5.254   192.168.5.254   516         0x8000000a 0x003933 2
192.168.6.254   192.168.6.254   515         0x8000000a 0x005e09 2
192.168.1.254   192.168.1.254   514         0x8000000a 0x00a4db 2

                Net Link States (Area 0)
Link ID         ADV Router      Age         Seq#       Checksum
172.16.4.1      192.168.4.254   521         0x80000007 0x00e30f
172.16.2.1      192.168.2.254   520         0x80000007 0x00f503
172.16.3.1      192.168.3.254   516         0x80000007 0x00ec09
172.16.5.1      192.168.5.254   516         0x80000007 0x00d916
172.16.6.1      192.168.6.254   515         0x80000007 0x00d01c
172.16.1.1      192.168.1.254   514         0x80000007 0x00fefc
CentralRT#
```

Figure 10: OSPF Link-State Database Summary

1.4 Infrastructure Configuration Details

1.4.1 Server and Central Router

The HTTP server is statically addressed to provide consistent access for the branch PCs.

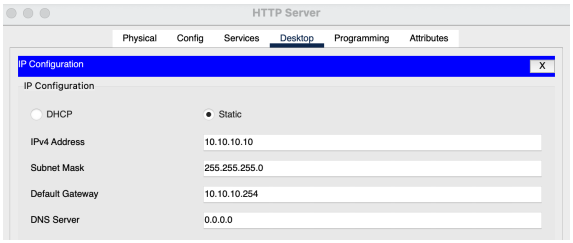


Figure 11: HTTP Server IP Setup

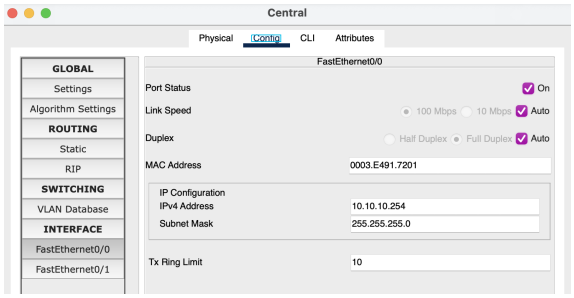


Figure 12: Central Router Fa0/0 Setup

1.4.2 VLAN Implementation

The Central Switch was configured with 6 VLANs to segregate traffic across the core.

```
Switch0(config-vlan)#interface fa0/1
Switch0(config-if)#switchport mode access
Switch0(config-if)#! remains in VLAN 1 (default)
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#interface fa0/2
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 2
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#interface fa0/3
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 3
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#interface fa0/4
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 4
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#interface fa0/5
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 5
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#interface fa0/6
Switch0(config-if)#switchport mode access
Switch0(config-if)#switchport access vlan 6
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#end
Switch0#write memory
%SYS-5-CONFIG_I: Configured from console by console

Switch0#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Gig0/1, Gig0/2
2    vlan2                            active    Fa0/2
3    vlan3                            active    Fa0/3
4    vlan4                            active    Fa0/4
5    vlan5                            active    Fa0/5
6    vlan6                            active    Fa0/6
```

Figure 13: VLAN Port Assignment and Summary

1.4.3 DHCP Server Fleet

Six DHCP servers provide dynamic addressing for the right-most PCs in each LAN.

Server	Server IP	Gateway	DNS	DHCP Pool Start
Server 1	192.168.1.253	192.168.1.254	192.168.1.253	192.168.1.1
Server 2	192.168.2.253	192.168.2.254	192.168.2.253	192.168.2.1
Server 3	192.168.3.253	192.168.3.254	192.168.3.253	192.168.3.1
Server 4	192.168.4.253	192.168.4.254	192.168.4.253	192.168.4.1
Server 5	192.168.5.253	192.168.5.254	192.168.5.253	192.168.5.1
Server 6	192.168.6.253	192.168.6.254	192.168.6.253	192.168.6.1

Table 1: Consolidated DHCP Server Parameters

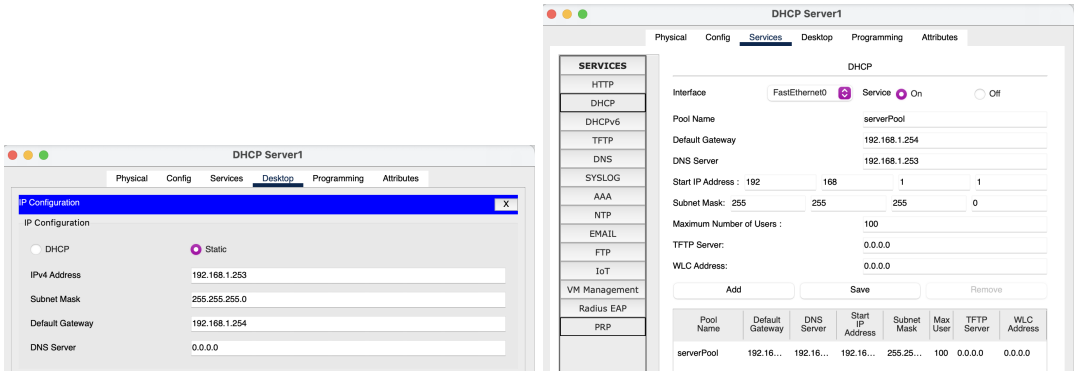


Figure 14: DHCP Server 1: IP and Service Configuration

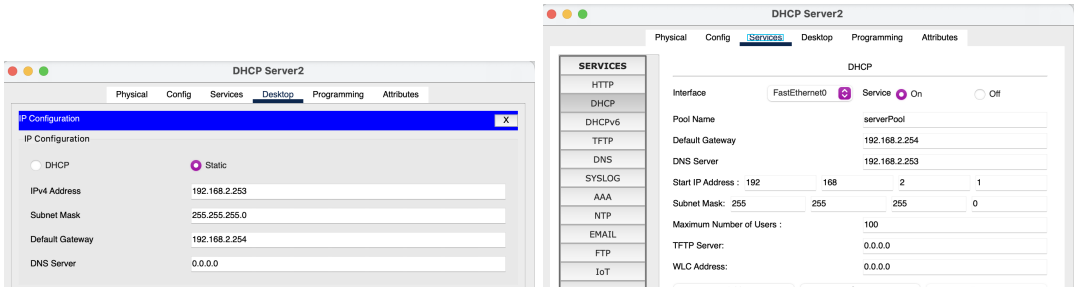


Figure 15: DHCP Server 2: IP and Service Configuration

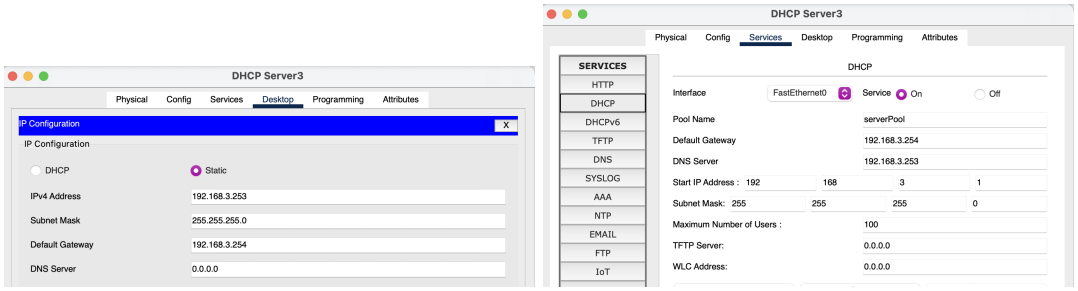


Figure 16: DHCP Server 3: IP and Service Configuration

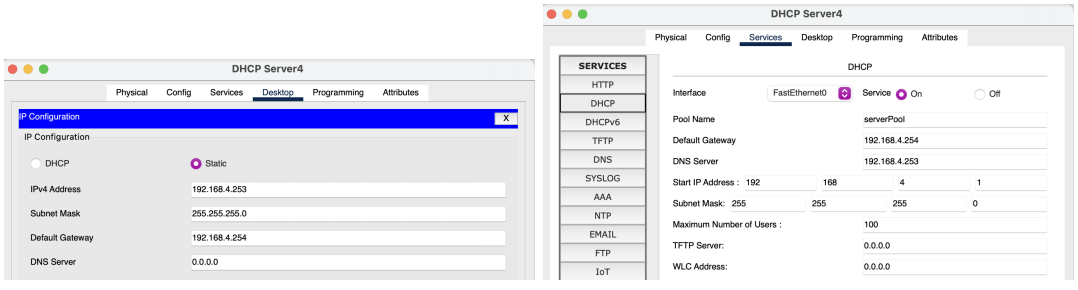


Figure 17: DHCP Server 4: IP and Service Configuration

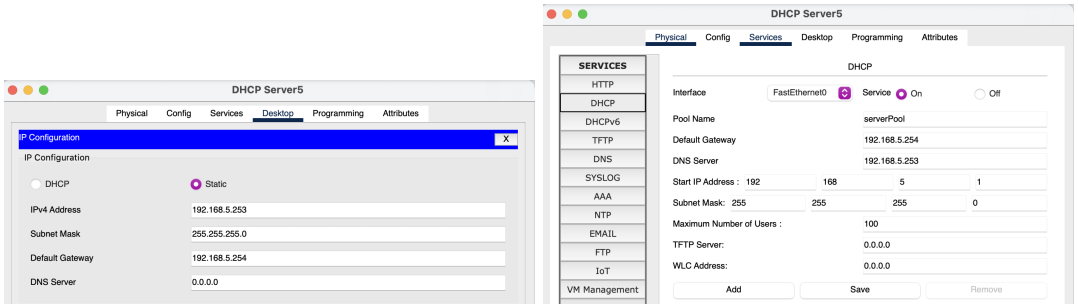


Figure 18: DHCP Server 5: IP and Service Configuration

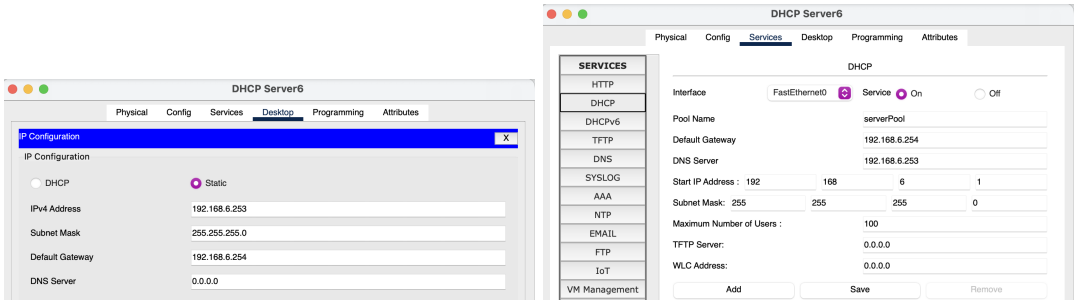


Figure 19: DHCP Server 6: IP and Service Configuration

1.4.4 Branch Router Interface Settings

Router	Interface	IPv4 Address	Subnet Mask
RT1	Fa0/0 (LAN)	192.168.1.254	255.255.255.0
	Fa0/1 (WAN)	172.16.1.1	255.255.255.248
RT2	Fa0/0 (LAN)	192.168.2.254	255.255.255.0
	Fa0/1 (WAN)	172.16.2.1	255.255.255.248
RT3	Fa0/0 (LAN)	192.168.3.254	255.255.255.0
	Fa0/1 (WAN)	172.16.3.1	255.255.255.248
RT4	Fa0/0 (LAN)	192.168.4.254	255.255.255.0
	Fa0/1 (WAN)	172.16.4.1	255.255.255.248
RT5	Fa0/0 (LAN)	192.168.5.254	255.255.255.0
	Fa0/1 (WAN)	172.16.5.1	255.255.255.248
RT6	Fa0/0 (LAN)	192.168.6.254	255.255.255.0
	Fa0/1 (WAN)	172.16.6.1	255.255.255.248

Table 2: Detailed Interface Assignments

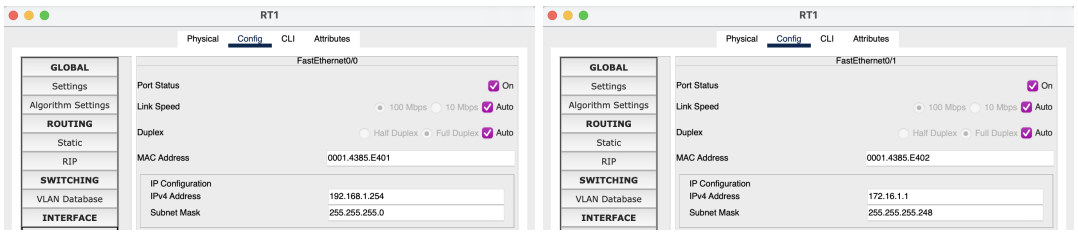


Figure 20: RT1 Interface Configurations

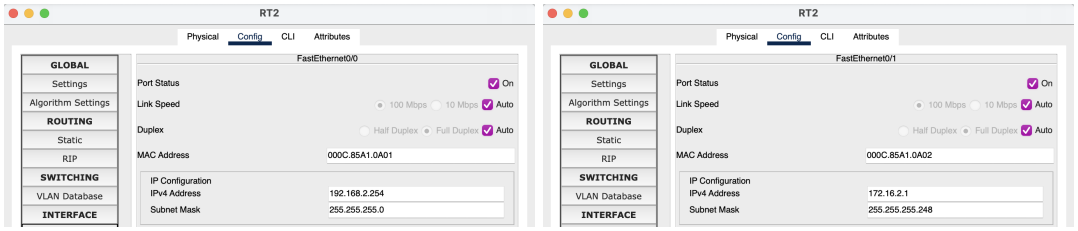


Figure 21: RT2 Interface Configurations

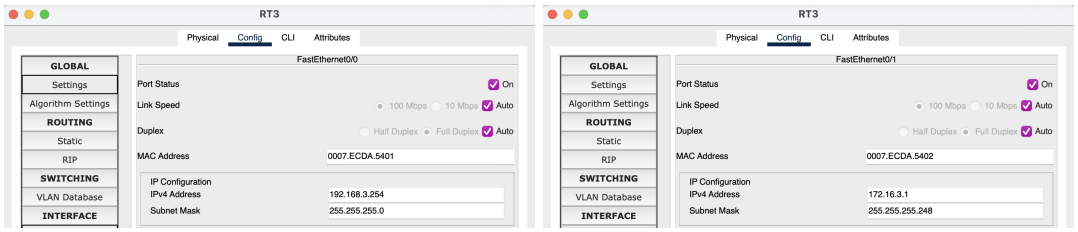


Figure 22: RT3 Interface Configurations

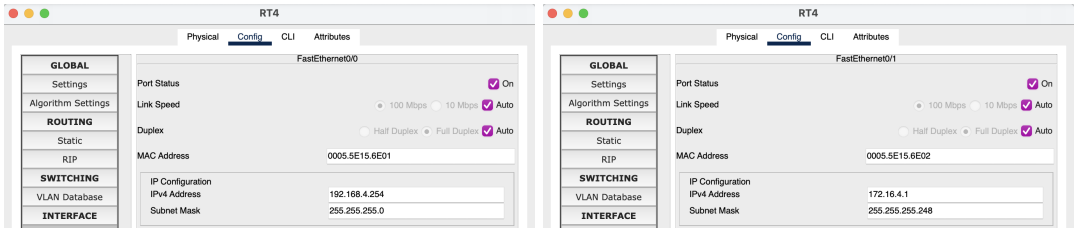


Figure 23: RT4 Interface Configurations

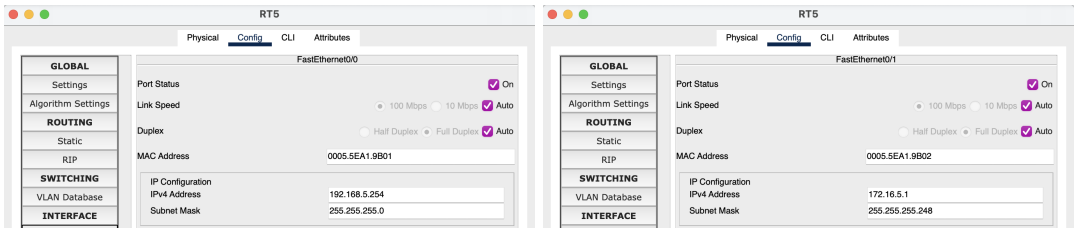


Figure 24: RT5 Interface Configurations

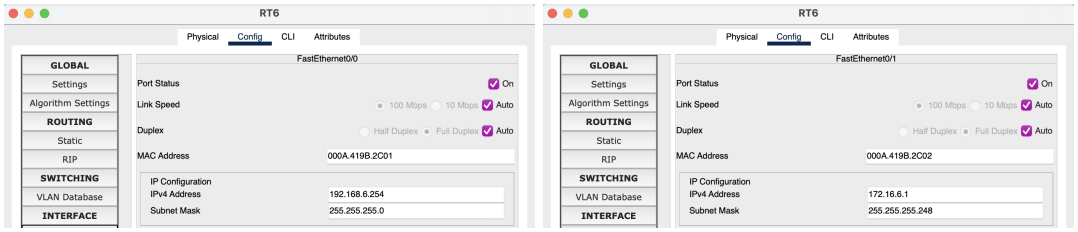


Figure 25: RT6 Interface Configurations

1.4.5 End-System (PC)s

The left-most PCs were configured statically, while the right-most PCs obtained addresses via DHCP.



Figure 26: PC Placement for Reference

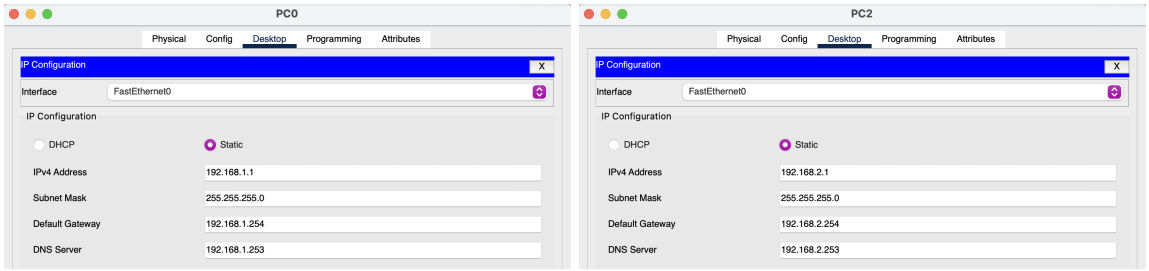


Figure 27: Static Config: PC 0 (left) and PC 2 (right)

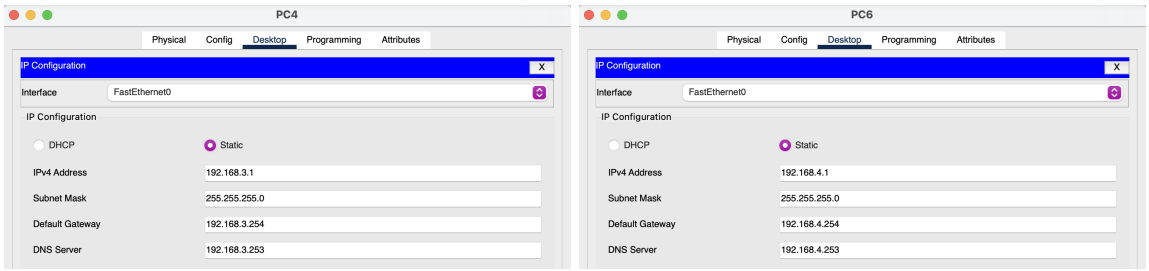


Figure 28: Static Config: PC 4 (left) and PC 6 (right)

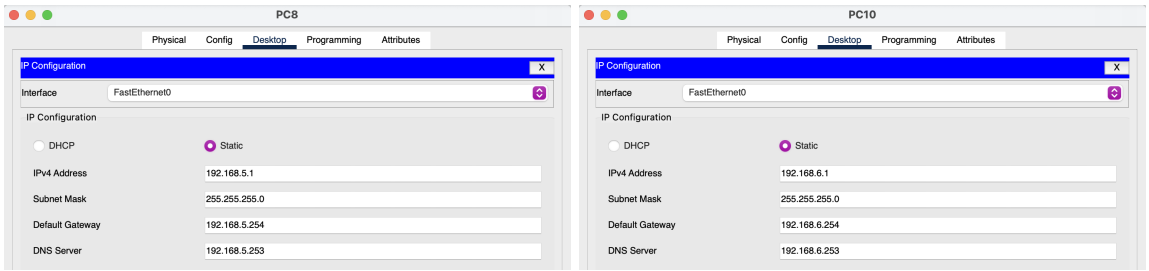


Figure 29: Static Config: PC 8 (left) and PC 10 (right)

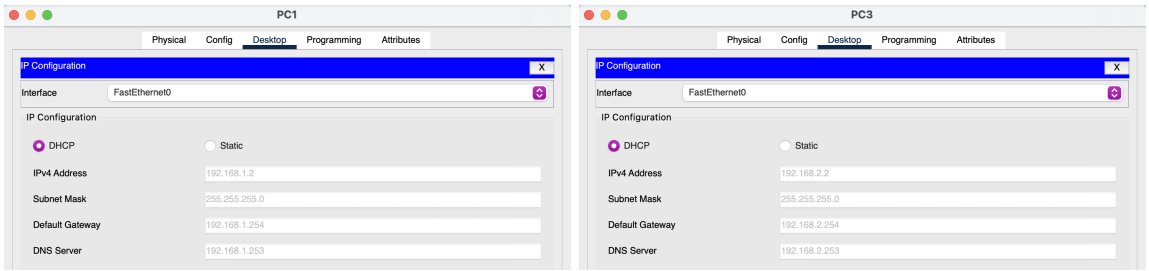


Figure 30: DHCP Config: PC 1 (left) and PC 3 (right)

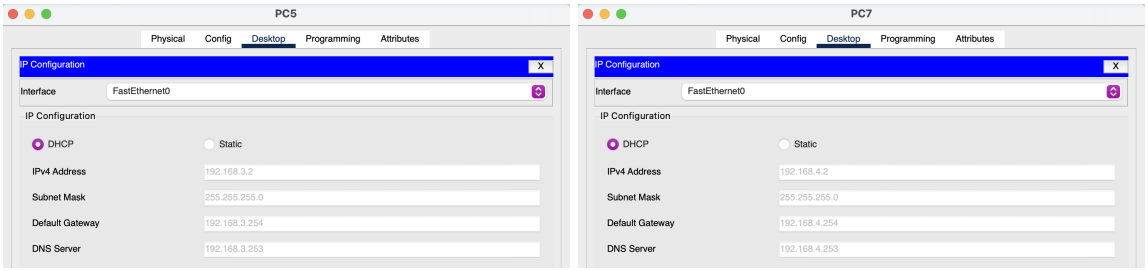


Figure 31: DHCP Config: PC 5 (left) and PC 7 (right)

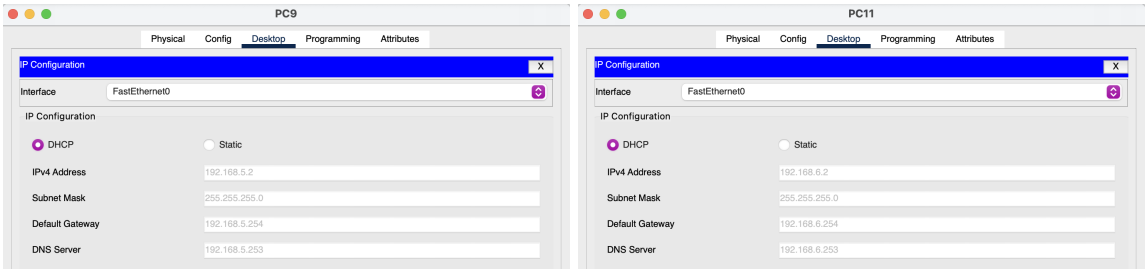


Figure 32: DHCP Config: PC 9 (left) and PC 11 (right)

1.5 Subnetwork Analysis

There are 13 total subnetworks in this architecture:

1. **LAN Subnets:** 2 LANs per edge router $\times$ 6 edge routers $= 12$ subnets.
2. **Management Subnet:** 1 subnet between the central router and the HTTP server.

Total: $12 + 1 = 13$ subnetworks.

2 Exercise 2: Inter-Domain Routing with BGP

2.1 AS Topology and IP Addressing

Five routers were added to represent distinct Autonomous Systems (AS). Connectivity is established via /30 subnets.

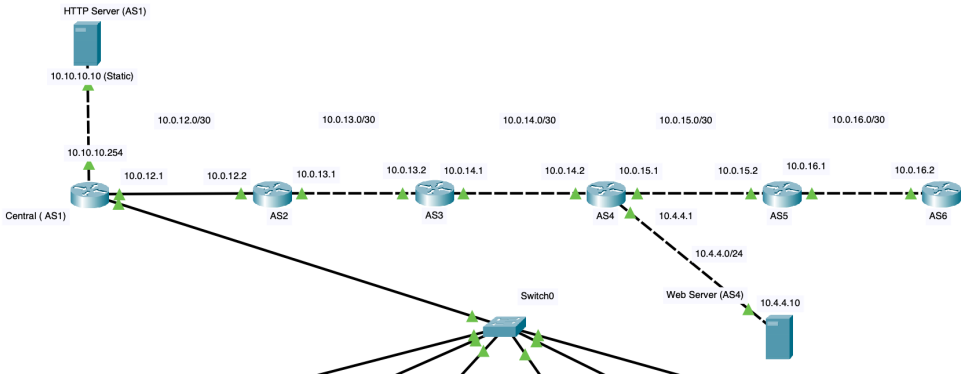


Figure 33: BGP Routing Topology Overview

- **Link Subnets:** 10.0.1x.0/30 where x corresponds to the AS number.
- **AS4 Internal:** Network 10.4.4.0/24 hosting a web server at 10.4.4.10.

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.254    YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES NVRAM  up                    up
FastEthernet0/1.1      172.16.1.6      YES NVRAM  up                    up
FastEthernet0/1.2      172.16.2.6      YES NVRAM  up                    up
FastEthernet0/1.3      172.16.3.6      YES NVRAM  up                    up
FastEthernet0/1.4      172.16.4.6      YES NVRAM  up                    up
FastEthernet0/1.5      172.16.5.6      YES NVRAM  up                    up
FastEthernet0/1.6      172.16.6.6      YES NVRAM  up                    up
FastEthernet0/0/0      unassigned      YES unset  down                  down
FastEthernet0/0/1      unassigned      YES unset  down                  down
FastEthernet0/0/2      unassigned      YES unset  down                  down
FastEthernet0/0/3      unassigned      YES unset  down                  down
FastEthernet0/1/0      unassigned      YES unset  down                  down
FastEthernet0/1/1      unassigned      YES unset  down                  down
FastEthernet0/1/2      unassigned      YES unset  down                  down
FastEthernet0/1/3      unassigned      YES unset  down                  down
Ethernet0/3/0          10.0.14.1       YES manual up                    up
Ethernet1/0            10.0.12.1       YES manual up                    up
Ethernet1/1            10.0.13.1       YES manual up                    up
Ethernet1/2            10.0.15.1       YES manual up                    up
Ethernet1/3            10.0.16.1       YES manual up                    up
```

Figure 34: AS1 Interface Addressing

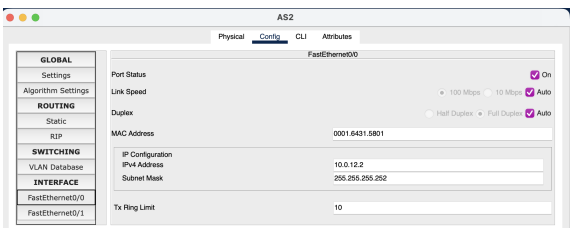


Figure 35: AS2 Config

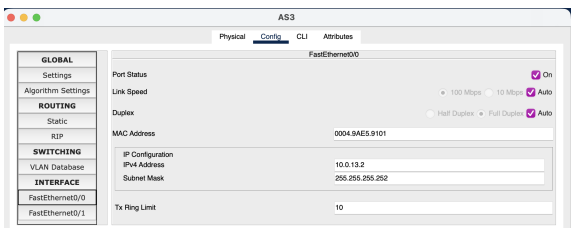


Figure 36: AS3 Config

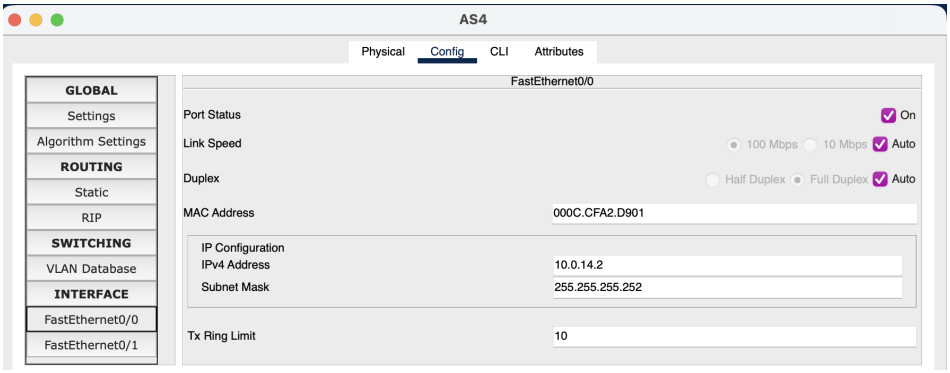


Figure 37: AS4 Interface (facing AS1)

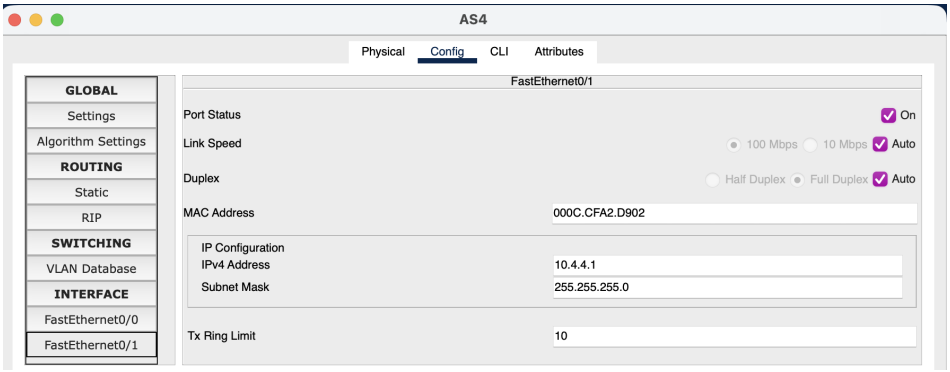


Figure 38: AS4 Interface (facing Web Server)

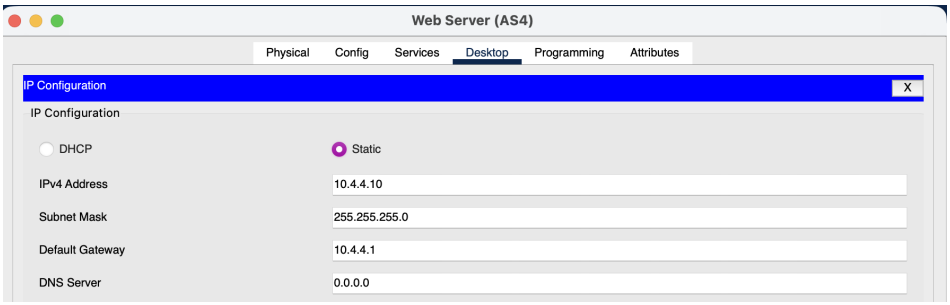


Figure 39: AS4 Web Server Setup

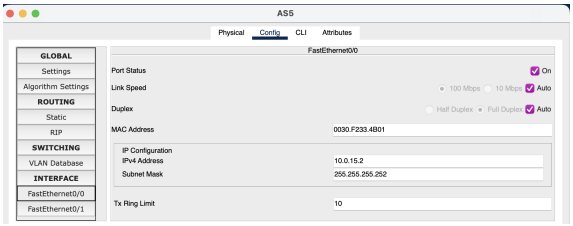


Figure 40: AS5 Config

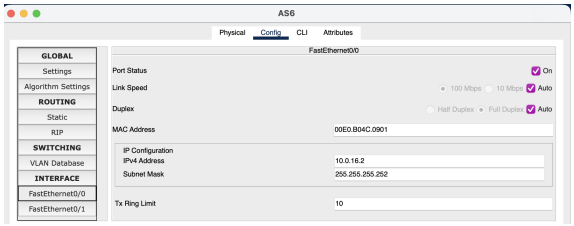


Figure 41: AS6 Config

2.2 BGP Protocol Configuration

2.2.1 Peer Configurations

The following screenshots detail the `router bgp` commands used to establish neighbor relationships and advertise local networks.

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.254    YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/1.1      172.16.1.6      YES NVRAM  up                    up
FastEthernet0/1.2      172.16.2.6      YES NVRAM  up                    up
FastEthernet0/1.3      172.16.3.6      YES NVRAM  up                    up
FastEthernet0/1.4      172.16.4.6      YES NVRAM  up                    up
FastEthernet0/1.5      172.16.5.6      YES NVRAM  up                    up
FastEthernet0/1.6      172.16.6.6      YES NVRAM  up                    up
FastEthernet0/0/0      unassigned      YES unset  down                  down
FastEthernet0/0/1      unassigned      YES unset  down                  down
FastEthernet0/0/2      unassigned      YES unset  down                  down
FastEthernet0/0/3      unassigned      YES unset  down                  down
FastEthernet0/1/0      unassigned      YES unset  down                  down
FastEthernet0/1/1      unassigned      YES unset  down                  down
FastEthernet0/1/2      unassigned      YES unset  down                  down
FastEthernet0/1/3      unassigned      YES unset  down                  down
Ethernet0/3/0          10.0.14.1       YES manual down                  down
Ethernet1/0            10.0.12.1       YES manual up                    up
```

```
AS2>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.12.2       YES NVRAM  up                    up
FastEthernet0/1        10.0.13.1       YES manual up                    up
Ethernet1/0            unassigned      YES unset  down                  down
Ethernet1/1            unassigned      YES unset  down                  down
Ethernet1/2            unassigned      YES unset  down                  down
Ethernet1/3            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

Figure 42: Interface Configurations for AS1 and AS2

```
AS3>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.14.1       YES manual up                    up
FastEthernet0/1        10.0.13.2       YES manual up                    up
Ethernet1/0            unassigned      YES manual down                  down
Ethernet1/1            unassigned      YES unset  down                  down
Ethernet1/2            unassigned      YES unset  down                  down
Ethernet1/3            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

```
AS4>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.14.2       YES manual up                    up
FastEthernet0/1        10.4.4.1        YES manual up                    up
Ethernet1/0            10.0.15.1       YES manual up                    up
Ethernet1/1            unassigned      YES unset  down                  down
Ethernet1/2            unassigned      YES unset  down                  down
Ethernet1/3            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

Figure 43: Interface Configurations for AS3 and AS4

```
AS5>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.15.2       YES manual up                    up
FastEthernet0/1        10.0.16.1       YES manual up                    up
Ethernet1/0            unassigned      YES unset  down                  down
Ethernet1/1            unassigned      YES unset  down                  down
Ethernet1/2            unassigned      YES unset  down                  down
Ethernet1/3            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

```
AS6>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.0.16.2       YES NVRAM  up                    up
FastEthernet0/1        unassigned      YES NVRAM  down                  down
Ethernet1/0            unassigned      YES unset  down                  down
Ethernet1/1            unassigned      YES unset  down                  down
Ethernet1/2            unassigned      YES unset  down                  down
Ethernet1/3            unassigned      YES unset  down                  down
Vlan1                  unassigned      YES NVRAM  administratively down down
```

Figure 44: Interface Configurations for AS5 and AS6

2.2.2 BGP Network Advertisements

To ensure reachability, the following specific network statements were applied in BGP configuration mode. (By using router bgp X) for ASX.

We advertised ASX's neighbors to be:

```
neighbor 10.0.1(X).1 remote-as (X-1)
neighbor 10.0.1(X+1).2 remote-as (X+1)
```

And it's reachable networks to be:

```
network 10.0.1(X).0 mask 255.255.255.252
network 10.0.1(X+1).0 mask 255.255.255.252
```

For AS1 (since it has its own subnet with its web server) we additionally added:

```
network 10.10.10.0 mask 255.255.255.0
```

Similarly, for AS4 (since it has its own subnet with its web server) we additionally added:

```
network 10.4.4.0 mask 255.255.255.0
```

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router bgp 1
Router(config-router)#neighbor 10.0.12.2 remote-as 2
Router(config-router)# network 10.10.10.0 mask 255.255.255.0
Router(config-router)# network 10.0.12.0 mask 255.255.255.252
Router(config-router)#exit
```

```
AS2#enable
AS2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
AS2(config)#router bgp 2
AS2(config-router)#neighbor 10.0.12.1 remote-as 1
AS2(config-router)#network 10.0.12.0 mask 255.255.255.252
AS2(config-router)#network 10.0.13.0 mask 255.255.255.252
AS2(config-router)#%BGP-5-ADJCHANGE: neighbor 10.0.12.1 Up

AS2(config-router)#neighbor 10.0.13.2 remote-as 3
```

Figure 45: BGP Neighbor/Network commands for AS1 and AS2

```
AS3#enable
AS3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
AS3(config)#router bgp 3
AS3(config-router)#neighbor 10.0.13.1 remote-as 2
AS3(config-router)#neighbor 10.0.14.2 remote-as 4
AS3(config-router)#network 10.0.13.0 mask 255.255.255.252
AS3(config-router)#network 10.0.14.0 mask 255.255.255.252
AS3(config-router)#
AS3(config-router)#%BGP-5-ADJCHANGE: neighbor 10.0.14.2 Up
```

```
AS4(config)#router bgp 4
AS4(config-router)# neighbor 10.0.14.1 remote-as 3
AS4(config-router)# neighbor 10.0.15.2 remote-as 5
AS4(config-router)# network 10.0.14.0 mask 255.255.255.252
AS4(config-router)# network 10.0.15.0 mask 255.255.255.252
AS4(config-router)# network 10.4.4.0 mask 255.255.255.0  #
                                                         ^
```

Figure 46: BGP Neighbor/Network commands for AS3 and AS4

```
AS5#enable
AS5#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
AS5(config)#router bgp 5
AS5(config-router)#neighbor 10.0.15.1 remote-as 4
AS5(config-router)#neighbor 10.0.16.2 remote-as 6
AS5(config-router)#network 10.0.15.0 mask 255.255.255.252
AS5(config-router)#network 10.0.16.0 mask 255.255.255.252
AS5(config-router)#%BGP-5-ADJCHANGE: neighbor 10.0.15.1 Up
```

```
AS6#enable
AS6#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
AS6(config)#router bgp 6
AS6(config-router)#neighbor 10.0.16.1 remote-as 5
AS6(config-router)#network 10.0.16.0 mask 255.255.255.252
AS6(config-router)#
```

Figure 47: CLI Network Statements: AS5 and AS6

2.3 Reachability Verification

Final connectivity was verified by performing a ping from AS1 to the Web Server located in AS4.

```
Router#ping 10.4.4.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.4.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/12/26 ms

Router#
```

Figure 48: Successful Connectivity Test: AS1 to AS4 Web Server